

# MTE 204 – Numerical Methods

## Tutorial #2 Questions

### Question 1:

Use Cramer's Rule to solve the following set of equations:

$$2x_2 + 5x_3 = 9$$

$$2x_1 + x_2 + x_3 = 9$$

$$3x_1 + x_2 = 10$$

Find the roots then substitute the values back into the equations to check your results.

### Question 2:

Use Gauss elimination to solve the following set of equations:

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

Find the roots then substitute the values back into the equations to check your results.

### Question 3:

Use Gauss elimination with partial pivoting to solve the following set of equations:

$$2x_1 - 6x_2 - x_3 = -38$$

$$-3x_1 - x_2 + 7x_3 = -34$$

$$-8x_1 + x_2 - 2x_3 = -20$$

Find the roots then substitute the values back into the equations to check your results.